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Research Paper

Rapid prediction of cervical cancer and high-grade precursor lesions: An integrated approach using low-field ^{1}H NMR and chemometric analysis

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ABSTRACT

Cervical cancer (CC) is a significant cause of morbidity and mortality in women, often preceded by high-grade cervical intraepithelial lesions (HSIL). Although conventional cytology (Pap smear) is widely used for screening, its sensitivity limitations and high false-positive rate reinforce the need for complementary methods. This study investigated the feasibility of low-field ^{1}H NMR spectroscopy combined with chemometric modeling to differentiate healthy individuals (CON) from patients with HSIL and CC. Principal Component Analysis (PCA) was applied to explore metabolic patterns and identify relevant spectral variables in group differentiation. PCA1 highlighted the separation between CC and the other groups, while PCA2 and PCA3 evidenced intermediate metabolic characteristics in HSIL, reinforcing its role as a transition stage. Three classification scenarios were evaluated using Data-Driven Soft Independent Modeling of Class Analogy (DD-SIMCA): (1) CON as target class, HSIL/CC as outclass classes; (2) HSIL as target class, CON as outclass class; and (3) CC as target class, CON as outclass class. Calibration was optimal (100 % SEN, SPE, ACC, MCC), and prediction showed higher efficacy in detecting CC (SPE = 100 %, MCC = 70 %), indicating that the model was more efficient in screening cervical cancer cases. Furthermore, low-field ^{1}H NMR has demonstrated potential as a metabolomic screening tool. It is a promising alternative due to its greater accessibility, lower operational cost, and non-invasive nature, complementing traditional methods of early detection of tumors and cervical lesions.

1. Introduction

Cervical cancer (CC) is the fourth most common neoplasm in terms of incidence and mortality in the female population worldwide [1]. Typically, CC is preceded by precursor lesions known as cervical intraepithelial lesions (SIL), with stages II and III classified as high-grade (HSIL) [2,3]. Early detection and treatment of these lesions can prevent their progression to CC [4]. Currently, cytological screening (Pap

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smear) is the most widespread test for screening for SIL and CC [5,6], but it has important limitations, such as low sensitivity and significant false-positive rates [6,7]. This reality highlights the need for complementary or alternative methods that are more assertive and have better diagnostic potential.

In this context, metabolomics associated with nuclear magnetic resonance (NMR) spectroscopy has been widely used to investigate the metabolome in several biofluids, including plasma and serum, with a focus on identifying biomarkers and unraveling biochemical processes associated with diseases such as cancer [8–10]. High-field (HF) NMR, in particular, has a long tradition in metabolomic applications [11–13], standing out for its high reproducibility and stability, rapid analysis, and simple, non-destructive sample preparation [14,15]. However, the high cost of HF spectrometers, their dependence on cryogenic systems, their large dimensions, and the need for highly specialized operators limit their access in many institutions [14,16,17].

Therefore, low-frequency (LF) NMR has become more affordable for exploring metabolic profiles [18]. Benchtop equipment has lower operating costs, does not require cryogenics, and facilitates maintenance, allowing laboratories of different sizes to access the technology. Although sensitivity and resolution are lower than HF's, LF NMR has proven suitable for detecting specific metabolites and metabolic fingerprinting in biofluids [17,19]. This technique has already been used to distinguish healthy individuals from people with type 2 diabetes from plasma samples [20] and early detection of sepsis in premature newborns through urinary profiles [21]. However, the application of this analytical tool in the study of neoplasms, especially cervical cancer, is still little explored.

Chemometric tools are essential to overcome the limitations inherent in LF NMR and extract the maximum information from the spectra. Methods such as principal component analysis (PCA) and independent modeling-based classification (SIMCA) have already demonstrated effectiveness in discriminating different groups in metabolomic studies [22,23]. Recently, single-class approaches, such as DD-SIMCA (Data-Driven Soft Independent Modeling of Class Analogy), have gained relevance by allowing the construction of robust models that describe only the target class (e.g., healthy individuals), evaluating the relevance of new samples to this same group [24,25]. This strategy is especially advantageous when more abundant data are available from a single class of interest or when the focus is on detecting metabolic deviations from a normal pattern [26–28].

Therefore, this study aimed to evaluate the efficiency of benchtop ^{1}H NMR spectral data from LF, combined with chemometric analyses, which can be promisingly employed in predicting CC and its high-grade precursor lesions (HSIL). For this purpose, an exploratory approach using PCA and a supervised approach through DD-SIMCA were applied to investigate the spectral information of LF accurately, providing classification models to differentiate diseased and healthy groups through a less invasive approach compared to conventional methods such as the Pap smear, evidencing its clinical potential for screening and diagnostic support and aiming at the development of an accessible, reliable and innovative screening tool in the detection of cervical tumors and lesions.

2. Methodology

2.1. Study population

This prospective descriptive study was approved by Ethics Committee of the Oncology Control Foundation Center of the Amazonas State (CEP/FCECON: 57108122.5.0000.0004-CAAE). Study participants were divided into three categories: CON – healthy control, HSIL – high-grade cervical intraepithelial lesions and CC – cervical cancer. Seventy-five women were selected to compose the DD-SIMCA model (twenty-five patients to compose each group). For model validation, thirty women were selected (fifteen healthy participants and fifteen with HSIL). All volunteer participants met the following criteria: (i) Women between 18 and 85 years of age, who presented a histopathological diagnosis compatible with CIN II/CIN III or invasive squamous cell carcinoma; (ii) Women who are virgins to cervical treatment; (iii) Women who have had two consecutive annual negative screening tests, only for control participants. All patients gave written informed consent prior to study admission.

2.2. Sample collection and preparation

Blood samples were collected by peripheral puncture using a vacuum collection system. This study collected 4 mL of blood in a collection tube containing the anticoagulant dipotassium ethylenediaminetetraacetic acid (EDTA) to obtain blood plasma. These samples were acquired during the routine of these patients, being performed in the triage room of the Basic Health Unit (for CON) and in the outpatient/inpatient clinic of the FCECON oncology hospital (for HSIL and CC). After collection, the blood samples were centrifuged at 4,500 rpm for 10 min at room temperature, and the supernatant obtained (plasma) was collected and stored in the freezer at −80 °C until analysis by Low-field NMR. For LF NMR spectroscopy analysis, plasma samples were thawed on ice and vortexed for a few seconds. Then, the samples were added to the 5 mm NMR tube without pretreatment.

2.3. Low-field NMR instrumentation and spectra acquisition

The ^{1}H spectra were acquired using a Spinsolve Multi X benchtop spectrometer (Magritek GmbH) equipped with permanent magnets, capable of generating a homogeneous magnetic field of 1.5 T (corresponding to a 60 MHz for the Proton and 15 MHz for the Carbon Larmor frequency). The system included a Spinsolve autosampler with a capacity of 20 tubes. For spectral acquisition, 500 μL of sample was directly inserted into a 5 mm NMR tube without solvents or deuterated solvents. Before the acquisition, the samples were placed in a thermostatic oven for 10 min to stabilize the temperature between 26.5 and 28.5 °C, corresponding to the probe temperature. The ^{1}H spectra were acquired in the "1D PROTON CDEC" mode by applying a 90° radio-frequency pulse. Sixty four scans were collected, with a total acquisition time of 14 min, an acquisition time of 3.2 s, and a repetition time of 15 s between each scan. In addition, the ^{13}C decoupling option was enabled. The bandwidth was set to 328 ppm, with a center frequency of 110 ppm. Nuclear Overhauser effect (NOE) enhancement and ^{13}C decoupling options were enabled. Additionally, the instrument was automatically shimmed every 30 min of analysis using a 10/90 (v/v) H_2O/D_2O mixture.

2.4. Multivariate analysis

All spectral pre-treatment, multivariate analysis, and post-processing procedures were conducted using Matlab 2021a® (The MathWorks Inc., Natick, USA) in conjunction with a user-friendly graphical interface named Grupo de Análise Multivariada em Matrizes Alimentares (GAMMA-GUI), which is available for free download on GitHub at https://github.com/appGAMMA. Before initiating any pre-treatment or multivariate analysis protocols, the spectral data were converted into numerical matrices within the Matlab environment.

2.4.1. Spectral pre-treatment

The free induction decay signals (FIDs) were processed with baseline and zero-order phase corrections. Spectral alignment was performed using the water peak as a reference at δ 4.74 ppm; however, this peak was excluded from the chemometric modeling. The spectral range and frequency filter settings were predefined to remove unwanted regions and noise, with a spectral range established between δ 0.00 and δ 8.00 ppm and a 0.5 Hz filter applied.

To address potential spectral or experimental interferences, three

normalization approaches were tested: maximum normalization (∞ norm), sum of absolute values (1 norm), and Euclidean normalization (2 norm). All spectral transformations and processing steps were carried out using a custom routine developed in MATLAB 2023b® (The MathWorks Inc., Natick, USA), ensuring consistency and precision in the analyses.

2.4.2. Principal component analysis (PCA)

PCA was used to examine the data's natural behavior. This method reduces the dimensionality of the dataset by using linear combinations of the original variables to maximize variance in the multivariate space. The new variables resulting from PCA are called principal components (PCs), which are orthogonal projections of each other. They can be obtained through the Singular Value Decomposition (SVD) of the matrix **X**, producing the score (**T**) and loading (**L**) matrices, as well as the residual matrix (**E**) for a given set of *A* PCs, **Eq. (1)**.

$$\mathbf{X} = \sum_{a=1}^{A} \mathbf{t}_a \mathbf{l}_a^{\mathrm{T}} + \mathbf{e} = \mathbf{T}_A \mathbf{L}_A^{\mathrm{T}} + \mathbf{E} \tag{1}$$

PCA aims to identify patterns and simplify data visualization while preserving as much of the total variance of the original dataset as possible. Principal components represent directions in the data space that explain the most significant variance, allowing for a more precise and focused analysis of the relationships between samples and their characteristics in a reduced-dimensional space [29].

2.4.3. Data-Driven–Soft independent modeling of class Analogy (DD-SIMCA)

DD-SIMCA [30] is an adaptation of SIMCA [31] based on the PCA decomposition of the training matrix X (I × J), which contains the pre-processed average spectra of I pepper samples described by J variables. **(Eq. (1)**. After the PCA decomposition of the training samples of the target class gives the global scores matrix **T**, loadings **P**, and residuals **E** for a given number of *A* PCs. Then the next step is started, in which two distances are obtained for each sample, the score distance (SD), h_{i}, and the orthogonal distance (OD), v_{i}, estimated for each training sample. The SDs and ODs represent population statistics, and critical membership levels can be established for their values. When a new sample is considered, it can be projected onto the model, and its SD and OD values are compared with the known critical levels to decide the class's membership [32].

SD, or leverage, is the squared Mahalanobis distance calculated from the model center to sample *i* within the score subspace given by **(Eq. (2)**, while OD, or residual variance, is calculated as the sum of the squared residuals for *A* PCs used in the PCA model given by **Eq. (3)**.

$$h_{\mathrm{i}} = \mathbf{t}_{\mathrm{i}}^{\mathrm{T}} \left(\mathbf{T}^{\mathrm{T}}\mathbf{T}\right)^{-1} \mathbf{t}_{\mathrm{i}} \tag{2}$$

$$v_{\mathrm{i}} = \sum_{\mathrm{j}=1}^{\mathrm{J}} e_{\mathrm{ij}}^{2} \; ; \mathrm{i} = 1, \cdots, \mathrm{I} \tag{3}$$

T is the global score matrix for *A* PCs, $\mathbf{t}_{\mathrm{i}}$ is the score vector for *i*th sample considering *A* PCs, and e_{ij} is an element from the concatenated residual matrix **E**. Then, the total distance for a sample is calculated by **Eq. (4)** as a weighted sum of the score distance, *h*, and the orthogonal distance, *v*.

$$c = N_h \frac{h}{h_o} + N_v \frac{v}{v_o} \propto \chi^2(N_h + N_v) \tag{4}$$

The parameters h_0 and v_0 are the estimated mean values for *h* and *v*, while N_h and N_v are the degrees of freedom (DoF). Initially, these parameters are unknown *a priori*, but they are calculated by a data-driven approach using the training data set [28],[32–34]. The DoF can be estimated, considering that the scores follow a χ^2-distribution, using the method of moments (classical) or the robust method. For more detail, the readers can see Ref. [32].

The threshold limit value for the target class is defined considering type I errors (α). For a specific α, the acceptance area is determined by **Equation (5)**, where χ^{-2} is the quantile of the χ^2-distribution with $N_h + N_v$ degrees of freedom (DoF). The model is now ready for classifying new samples, with the acceptance plot visualizing the acceptance area in terms of score and orthogonal distances. Each sample from the training set is characterized by its position in these new coordinates. A sample can be classified as part of the target class if its distance is within the acceptance area, as extreme outside the area, or as an outlier (non-member). For a given α, the acceptance area is set, and any new sample's distance is compared to the critical threshold $c_{crit}(\alpha)$. Samples with distances below this threshold are accepted as part of the target class, while those exceeding the threshold are classified as extreme or outliers [2,32–34].

$$c \leq c_{crit}(\alpha) = \chi^{-2}(1 - \alpha, N_h + N_v) \tag{5}$$

Finally, the test samples can be classified using the model, and the results are presented in the acceptance plot, which can be given in type I error α and II error β [16,27,28,33].

2.4.4. Model performance

To ensure the robustness of the results and the reliability of the DD-SIMCA models, leave-one-out cross-validation (LOOCV) was used. This method allows the model's performance to be evaluated without relying on a specific sample, using each sample individually for validation while the other samples are used for training. The performance results, including sensitivity, specificity, and global efficiency (EFI), were evaluated throughout the cross-validation steps. Three specific scenarios were assessed, in which different classes were used as the target for evaluation:

Scenario 1: CON as the target class, with HSIL and CC as the external classes.

Scenario 2: HSIL as the target class, with CON as the external class.

Scenario 3: CC as the target class, with CON as the external class.

For each scenario, the best values of sensitivity and specificity from cross-validation were used as metrics to select the most appropriate model. Furthermore, the Matthews correlation coefficient (MCC) was used to provide a balanced assessment of the test performance, considering both true positives and true negatives. To validate the generalization of the models, an external dataset was also used. This additional external validation step allowed the models' ability to generalize to new data to be evaluated, ensuring that the results were not specific to the original dataset. External validation strengthened the reliability of the models and extended their applicability to different scenarios.

For the model evaluation on the test set, metrics such as accuracy (ACC), MCC, sensitivity (SEN), and specificity (SPE) were used. These metrics were calculated based on the values of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), providing a comprehensive evaluation of the reliability and accuracy of the model classifications, considering different significance level values α (0.01 and 0.05) **Equations (6) to (9)**."

$$SEN = \frac{TP}{TP + FN} \tag{6}$$

$$SPE = \frac{TN}{TN + FP} \tag{7}$$

$$ACC = \frac{TN + TP}{TN + TP + FN + FP} \tag{8}$$

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{9}$$

3. Results and discussion

3.1. Characteristics of study participants

Participants were women aged 19–76 years, with mean ages of 38.92 years for the control group (CON), 40.32 years for the pre-cancer group (HSIL), and 43.56 years for the cervical cancer group (CC), as summarized in Table 1. Several variables potentially influencing the metabolic profiles detected by ^{1}H NMR were recorded, including menstrual cycle phase, 75 % women of reproductive age in the CON and HSIL groups, and 64 % in the CC group, versus 10, 20, and 30 % postmenopausal women in the CON, HSIL, and CC groups, respectively; use of hormonal contraceptives (45 % in controls, compared with 60 % and 64 % in the HSIL and CC groups); and HPV status, which was negative in the control group and positive in 93 % and 96 % of HSIL and CC cases, respectively. Furthermore, smoking habits remained constant between groups (7.5 %), and dietary patterns revealed that 10 % of participants in the CON and CC groups adhered to a balanced diet, compared with 22 % in the HSIL group.

The inclusion of these critical variables not only allows for adjustment of potential confounders, thus increasing the robustness of intergroup comparisons, but also enriches the dataset by incorporating realistic clinical variability. This heterogeneity, which mirrors the complexity commonly encountered in clinical practice, enhances the predictive power and generalizability of the chemometric models. As highlighted by Barros and coauthors (2021), factors such as early sexual debut, multiple sexual partners, smoking, contraceptive use, and sexually transmitted infections are significantly associated with the development of cervical lesions and tumors [35].

3.2. Visual inspection spectral graph

Fig. 1 shows the mean low-field ^{1}H NMR spectra of blood plasma from the three groups evaluated: healthy controls (CON), patients with precancerous lesions (HSIL), and patients with cervical cancer (CC). An increase in the intensity of signals attributed to lipid metabolites (LDL and VLDL), lactate, N-acetylated glycoproteins (NAG), citrate, choline, glycine, α-glucose, uridine, and unsaturated lipids is observed in the HSIL and CC groups compared to the CON group. The low-field NMR technique has spectral resolution limitations, which can complicate the separation of signals and compromise the accuracy of metabolite identification due to spectral overlap, especially when compounds have similar chemical structures. These limitations are inherent to the lower magnetic field intensity and lower resolution compared to high-field NMR, resulting in a reduced ability to distinguish closely spaced peaks. Baseline adjustments and phase correction were applied to minimize these challenges, enabling more accurate data interpretation. However, peak assignments were made based on available literature and the Human Metabolome Database (HMDB), allowing for a well-founded analysis but with potential margins of uncertainty in the exact discrimination of the compounds [36,37].

Changes in peak intensity in HSIL and CC may reflect changes in cellular metabolism associated with tumor proliferation and the inflammatory microenvironment [38,39]. In particular, the increase in lactate in CC suggests a predominance of aerobic glycolysis, which is compatible with the "Warburg effect," a phenomenon in which tumor cells undergo metabolic reprogramming to optimize glucose uptake and ATP production via glycolysis, resulting in the accumulation of lactate even under normoxic conditions, providing tumor cells with metabolic advantages favorable for growth and proliferation [40].

The increase in spectral intensity attributed to NAG glycoproteins in the HSIL and CC groups may be related to adaptive immune and inflammatory responses, processes frequently associated with neoplastic cells and tumor progression [41,42]. Furthermore, the increase in signals corresponding to LDL and VLDL may indicate more significant lipid mobilization, an essential process to meet tumor growth's structural and energetic demand [43].

The spectral signals of citrate, choline, and glycine, which are more intense in HSIL and CC, suggest alterations in intermediary metabolism and cell membrane biosynthesis, processes that are fundamental for the maintenance of cell proliferation [44]. Elevated citrate in CC may reflect the activation of metabolic pathways associated with lipid synthesis since this metabolite plays a fundamental role as a precursor for fatty acids in *de novo* lipogenesis [45,46]. Increased choline may be related to activation of the choline kinase pathway, a phenomenon frequently reported in solid tumors due to increased phosphatidylcholine synthesis, a principal constituent of cell membranes [47,48]. Similarly, elevated glycine may indicate increased nucleotide synthesis, which is essential for DNA replication and the accelerated cell proliferation observed in tumors [49,50].

Furthermore, increased α-glucose and uridine may be associated with increased glucose uptake and RNA synthesis, which play critical roles in tumor growth [51]. Increased glucose uptake by neoplastic cells has been widely documented in cancer metabolism studies, as it provides essential substrates for the biosynthesis of biomolecules and the maintenance of cell proliferation [51,52]. Similarly, the elevation of uridine may reflect an intensification of nucleotide synthesis and phosphoribosylation, mechanisms necessary for genome replication and cell survival under tumor stress conditions [51,52].

3.3. Exploratory analysis using PCA

PCA was applied to ^{1}H NMR data to explore metabolic differences in low-field ^{1}H NMR spectra obtained from the blood plasma of the three groups evaluated (CON, HSIL, and CC). The 3D score plot, presented in Fig. 2, reveals a partial separation between groups, with the first three principal components (PCs) explaining 96.9 % of the total variance in the data. This approach allowed us to identify metabolic patterns associated with the different stages of disease progression. PC1 explained most of the variance, indicating that it captures the most relevant metabolic alterations between groups. The additional components, PC2 (2.17 %) and PC3 (1.79 %) contributed to the complementary discrimination of metabolic differences. Despite the high variance explained by the first three PCs, it is important to consider that part of the metabolic information may be distributed in components with lower variance, requiring caution in interpreting differences between groups.

The score graph shows a partial separation between the three groups, suggesting metabolic differences in blood plasma. The CON group (green) presents a more homogeneous and isolated distribution in the three-dimensional space, which may indicate less metabolic variability among healthy individuals. On the other hand, the HSIL (yellow) and CC (red) groups are projected more closely, suggesting a more remarkable metabolic similarity between patients with precancerous lesions and those with cervical cancer when compared to the control group.

This proximity between HSIL and CC in three-dimensional space may be associated with the continuous progression of metabolic alterations

Table 1
General characteristics of study participants.

Variable	CON f (%)	HSIL f (%)	CC f (%)
Menstrual cycle phase			
Women of reproductive age	75	75	64
Postmenopausal women	10	20	30
Hormone use			
Hormonal contraceptives	45	60	64
HPV+	0	93	96
Smoking habit	7.5	7.5	7.5
Dietary habits			
Balanced diet	10	22	10

Where: f (%) = Relative Frequence; CON = healthy individuals; HSIL = precancerous lesions; CC = cervical cancer.

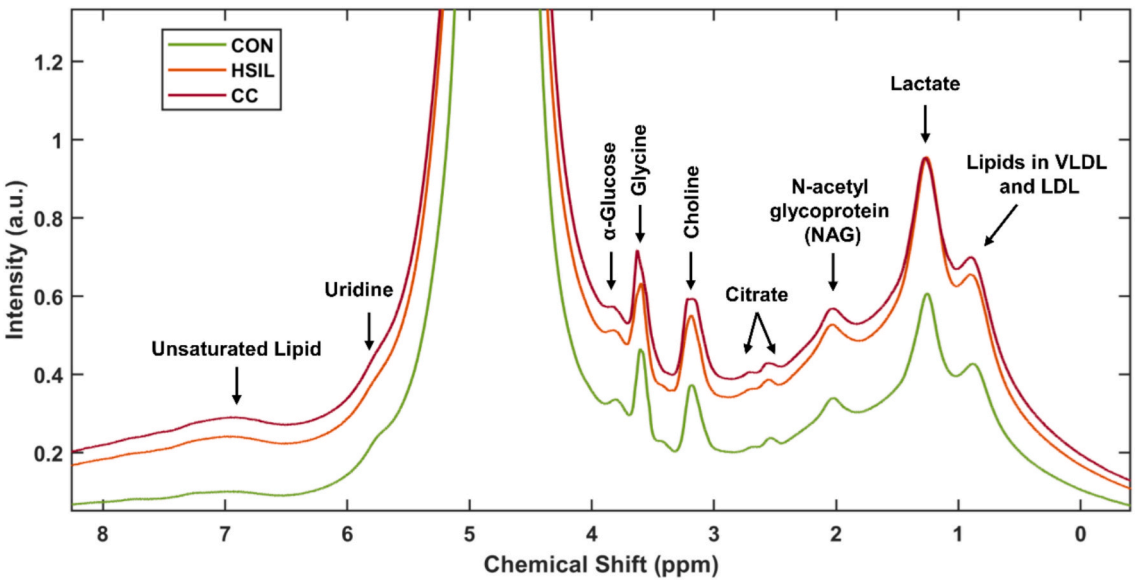


Fig. 1. Low-field ^{1}H NMR spectra of blood plasma from the groups: healthy controls (CON), patients with precancerous lesions (HSIL), and patients with invasive cervical cancer (CC). Peak identifications were performed based on the available literature and the Human Metabolome Database (HMDB).

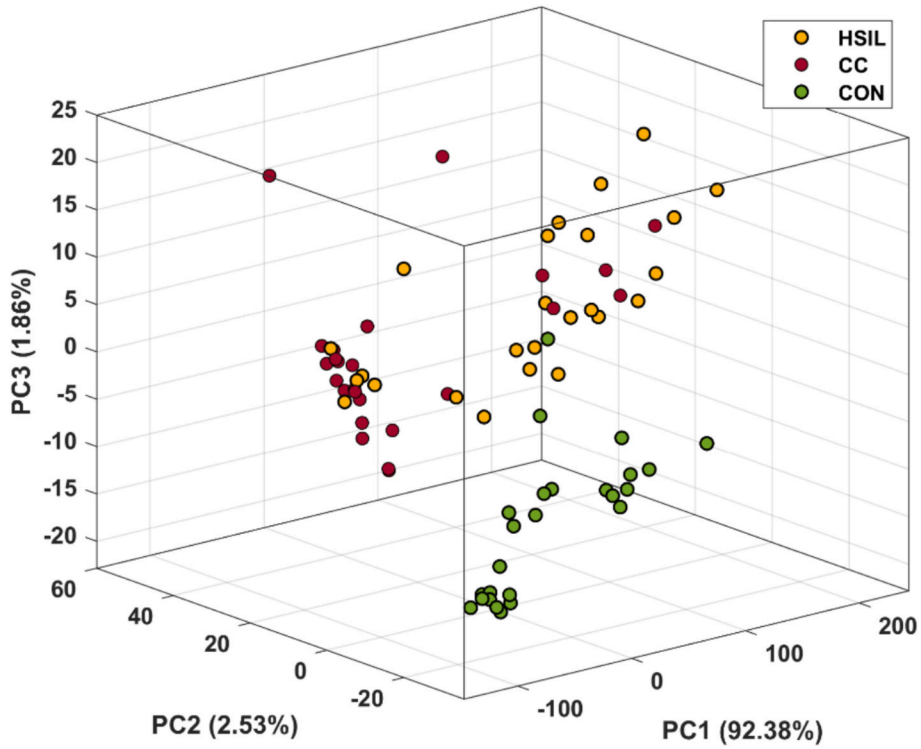


Fig. 2. Three-dimensional plots of PCA scores were applied to ^{1}H NMR spectra of blood plasma from three groups of patients: controls (CON, green), patients with precancerous lesions (HSIL, yellow), and patients with cervical cancer (CC, red).

throughout cancer development. The HSIL group represents an intermediate stage in which biochemical changes are already observed in plasma, including variations in the levels of lipids, amino acids, and metabolites of energy metabolism. Although these alterations are still less pronounced than in cervical cancer, they share common metabolic signatures with the CC group. This may justify the partial overlap between these groups in the multivariate analysis.

Additionally, the intermediate position of the HSIL cluster between CON and CC reinforces the hypothesis of a progressive metabolic transition. This suggests that, as the precancerous lesion evolves into cervical cancer, the metabolic changes become more intense and systematic, resulting in the displacement of the samples along the main components. The more excellent proximity of the HSIL and CC projections supports the idea that the metabolic characteristics of cervical cancer begin to manifest themselves even in the precancerous phase, which may have implications for the use of metabolic biomarkers in the early detection of the disease.

To better understand the metabolic variations between the analyzed groups, each principal component (PC) was evaluated individually, along with the loading plots, to identify which spectral variables

associated with the metabolites contributed most to the observed discrimination (Fig. 3). Although the use of more robust analytical techniques is necessary to accurately identify and confirm the metabolites, the analysis of the loading values allows us to suggest the most likely compounds, based on the initial assignment made in the spectrum. In this way, we are able to discuss and better understand the potential metabolic markers that could be associated with the behavior observed in the data, providing a solid foundation for interpreting the variations in the metabolic profiles, even without detailed experimental confirmation.

In PC1, it is observed that most samples from the HSIL and CON groups are projected in the positive quadrant, while samples from the CC group predominate in the negative quadrant (Fig. 3 **– I (A)**). As discussed previously, the metabolic proximity between HSIL and CON may reflect that HSIL represents an intermediate stage, sharing specific metabolic characteristics with the healthy state and exhibiting alterations associated with tumor progression [46,50]. The PC1 loadings graph (Fig. 3 **– I (B)**) highlights the most relevant spectral variables for separating the groups. The spectral region between 3.00 and 0.50 ppm was particularly significant, with several metabolites loading significantly (above the dashed red line). The primary metabolites involved were glycine (δ = 3.61 and 3.56 ppm), choline (δ = 3.19 and 3.14 ppm), N-acetyl glycoproteins (NAG, δ = 2.00 ppm), lactate (δ = 1.26 ppm) and lipids (δ = 0.96 ppm). Reinforcing its relevance in the differentiation of the groups, as seen in the differentiation of the spectra (Fig. 1).

The analysis of the PC2 scores (Figure II (A)) revealed a more dispersion of the samples along this component, with the HSIL group predominantly in the negative quadrant. In contrast, the CC and CON groups are distributed in opposite regions, mostly occupying the positive quadrant. This pattern suggests that PC2 captures more subtle metabolic

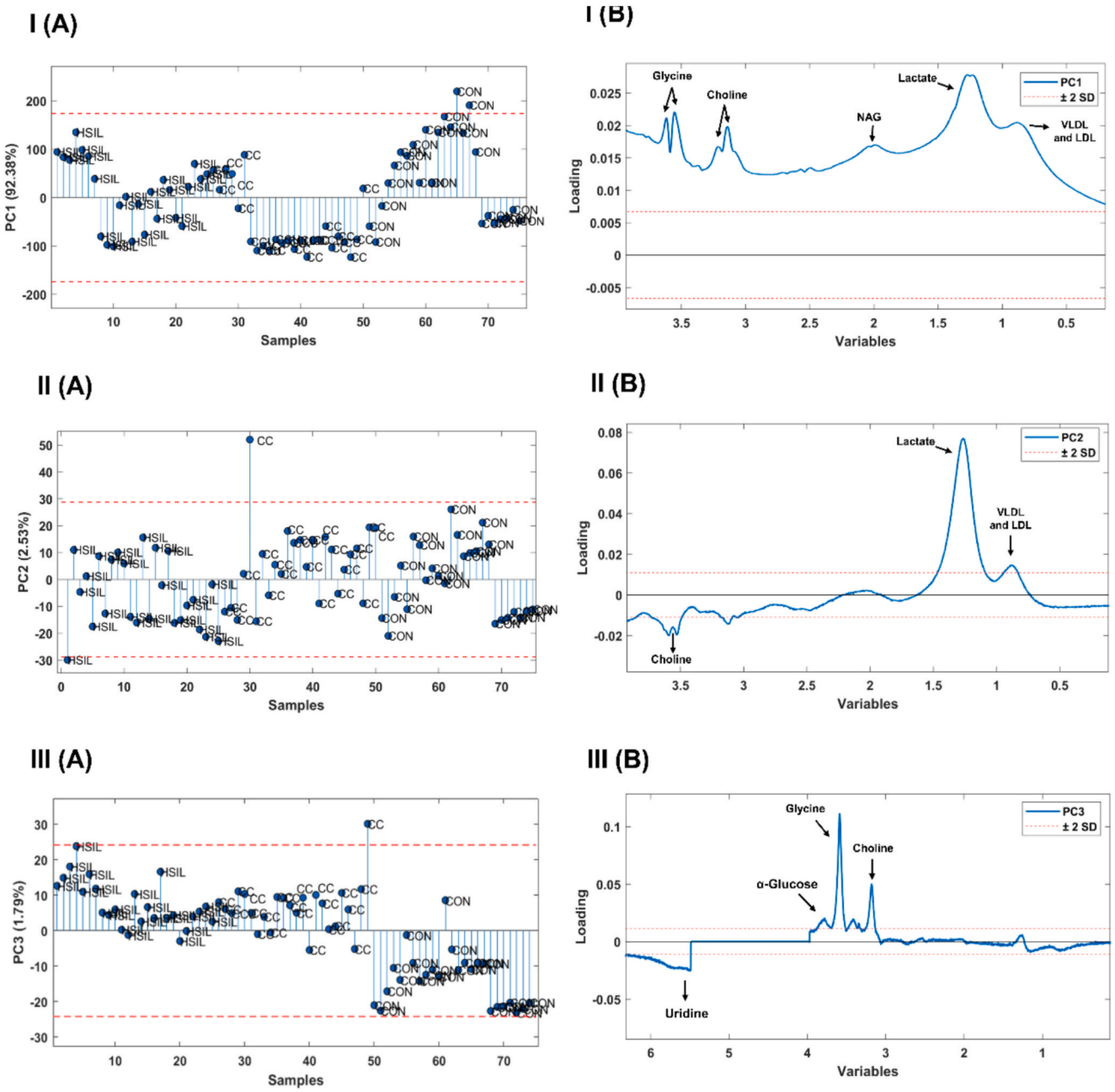


Fig. 3. Results of PCA applied to ^{1}H NMR spectra of blood plasma investigating the behavior of each PC individually, where: (I) Scores (A) and loadings (B) graphs for PC1, (II) Scores (A) and loadings (B) graphs for PC2, and (III) Scores (A) and loadings (B) graphs for PC3.

variations, complementing the discrimination observed in PC1, especially about the separation between HSIL and the other groups. The PC2 loading graphs indicate that only choline (δ = 3.19 and 3.14 ppm), lactate (δ = 1.26 ppm), and lipids associated with VLDL and LDL (δ = 0.96 ppm) presented significant loadings. Choline exhibited a negative loading, while lactate and lipids showed positive loadings. This suggests that samples from the HSIL group, with negative PC2 scores, have relatively lower choline levels. In contrast, CC and CON, which have positive scores, exhibit a relative increase in lactate and lipids. This behavior reinforces the role of choline in cell membrane remodeling, which is frequently altered in tumors, and highlights the importance of lipid metabolism in disease progression [40,43].

In the PC3 evaluation, a more evident separation between the groups was observed, with the CON group predominantly in the negative quadrant, while HSIL and CC occupied the positive quadrant (Fig. 3 **– III (A)**). This behavior suggests that PC3 captures specific metabolic patterns that distinguish healthy individuals from those with precancerous lesions and cervical cancer. The primary metabolites established in PC3 were glycine (δ = 3.61 and 3.56 ppm), choline (δ = 3.19 and 3.14 ppm), α-glucose (δ = 3.83 ppm) and uridine (δ = 5.80 ppm). Glycine and choline showed high positive loadings, indicating that their levels are relatively higher in the HSIL and CC groups. In contrast, uridine showed a negative loading, contributing more significantly to the CON group. This behavior may reflect the control group's maintenance of normal metabolic processes and lower inflammatory levels [49,50]. The α-glucose, with a positive charge, suggests an increase in the glycolytic rate in the HSIL and CC groups, in agreement with the metabolic changes associated with the Warburg effect in tumor cells [51,52].

3.4. Supervised analysis using DD-SIMCA

The DD-SIMCA approach was used to obtain more reliable results and a robust model for classifying samples using metabolic profiles through ^{1}H NMR data. This method was chosen because it is particularly suitable for single-class classification problems, where the main objective is not to separate distinct groups through a traditional supervised model but rather to build a statistical model based on a reference set (target class) and evaluate the similarity of new samples (outer classes) concerning this model [28,53].

In this study, we analyzed the application of the DD-SIMCA model in three different scenarios. The first scenario considers all classes (CON, HSIL, and CC), using the CON group as the reference class. This approach allows us to assess whether the metabolic profiles of HSIL and CC present statistically significant deviations from the controls, i.e., it investigates whether these classes differ from the healthy group. In the other two scenarios, we sought to simulate a situation closer to the real application of the method in clinical screening, where a patient arrives for the exam without prior knowledge about his/her condition. In this context, we defined HSIL and CC as the reference class and used external samples to simulate an individual's situation without a confirmed diagnosis. This approach reflects the method in metabolic screening, where the main objective is to identify whether a patient has a metabolic profile compatible with a precancerous lesion or cervical cancer, assisting in the decision to refer for more detailed diagnostic tests.

Fig. 4 **(A)** shows the influence of the number of PCs on key model performance indicators, including sensitivity (SEN), specificity (SPE), and overall efficiency (EFI). It is observed that using 1 to 3 PCs, the models show high sensitivity, ensuring the correct classification of most samples in the CON group but low specificity, which results in the incorrect acceptance of HSIL and CC samples and, consequently, an increase in false negatives. On the other hand, when using 3 to 7 PCs, a significant improvement in specificity and a reduction in the incorrect acceptance of HSIL and CC samples are observed, although there is a slight decrease in sensitivity and a small increase in false positives. Thus, the analysis of the FoMs (Figure of Merits) indicates that the overall efficiency reaches its peak between 5 and 7 PCs, making the configuration with 5 PCs ideal for building the models, as it maintains the high sensitivity necessary to correctly identify patients with metabolic profiles suggestive of injury while increasing specificity to reduce the misclassification of HSIL and CC samples as healthy.

Fig. 4 **(B)** shows the acceptance graph with the distribution of samples within the DD-SIMCA decision space, using logarithmic scales to

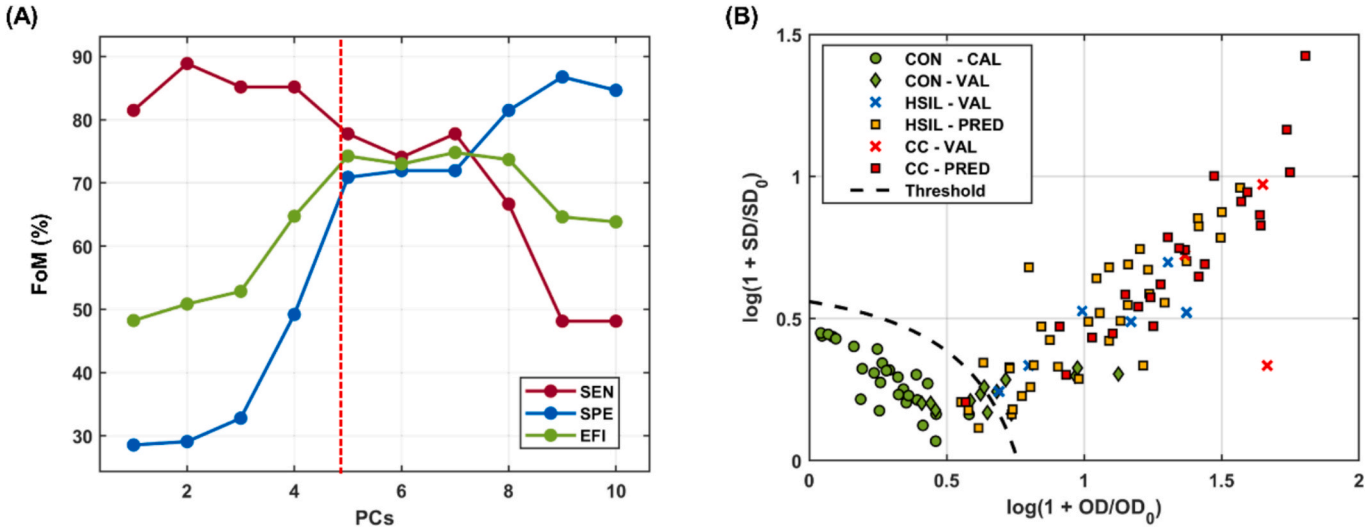


Fig. 4. Results of DD-SIMCA modeling for scenario 1. **(A):** Performance evaluation of the DD-SIMCA model, using the compliant method, as a function of the number of principal components (PCs). The indicators analyzed include sensitivity (SEN), specificity (SPE), and overall efficiency (EFI). The dashed red line indicates the number of PCs selected to build the final model. (B) Samples will be distributed in the DD-SIMCA decision space. The x-axis represents the normalized orthogonal distance ($\log(1 + OD/OD_0)$), and the y-axis represents the normalized score distance ($\log(1 + SD/SD_0)$). The acronyms represented in the graph indicate the different sets of samples used in the modeling and validation, where CON − CAL: Samples from the control group used for model calibration; CON − VAL: Samples from the control group used for model validation; HSIL − VAL: Samples from the HSIL group used in model validation; HSIL − PRED: Samples from the HSIL group used as external samples for prediction testing; CC − VAL: Samples from the CC group used in model validation; CC − PRED: Samples from the CC group used as external samples for prediction testing. The dotted black line represents the statistical decision threshold (α = 0.01), above which the samples are classified as outside the control model, indicating significant metabolic deviations.

normalize the statistical distances. The metrics used include the Orthogonal Distance (OD), which quantifies the deviation of each sample about the modeled PCA subspace, and the Score Distance (SD), which represents the relative position of the samples within this space [28]. Each symbol in the graph corresponds to one of the groups analyzed, and the black dotted line in the graph defines the statistical decision threshold ($\alpha = 0.01$), established to ensure that only 1 % of the CON samples are wrongly classified as "outside" the model due to normal statistical variations [27,53]. Analysis of the graph shows that most of the CON samples are below this threshold, evidencing the consistency of the model in correctly representing healthy individuals. In contrast, the samples from the HSIL and CC groups are predominantly distributed above the cutoff line, reflecting significant metabolic alterations that differ from normal patterns.

However, some samples from the HSIL group are close to the decision threshold, suggesting that this group presents more metabolic heterogeneity. This partial overlap between HSIL and CON corroborates what was previously observed in the PCA, indicating that specific HSIL samples have intermediate metabolic profiles and share characteristics with the CON and CC groups. This reinforces the need to improve the statistical modeling by expanding the sample base or by including new metabolic variables that can refine the separation between the groups.

The approach used in this study included applying external validation samples, i.e., data independent of the set used in the construction of the models, ensuring a more rigorous evaluation of DD-SIMCA performance. In Fig. 4 **(B)**, these external samples are shown, corresponding to individuals from the three groups in the validation (VAL) phase. Using these external samples is essential to test the model's generalization, as it allows us to verify whether it maintains its metabolic distinction capacity even when applied to previously unseen data. As previously observed, some HSIL-VAL samples are also close to the statistical threshold, reinforcing the metabolic heterogeneity of this group. This observation highlights the importance of a robust model capable of dealing with natural metabolic variations without compromising its ability to detect changes associated with disease development.

Fig. 5 presents the results of scenario 2, which simulates an actual clinical application in which the method is used to evaluate patients without a previous diagnosis, estimating their metabolic similarity with the target classes HSIL and CC. Unlike the initial approach, which used the CON group as a reference, these scenarios now configure HSIL and CC as the classes of interest, allowing the model to classify new patients based on statistical proximity to previously modeled metabolic profiles. This strategy is essential to validate the model's generalization capacity, minimizing overfitting and increasing its robustness for practical applications while reflecting the complexity inherent in real clinical scenarios, in which the correct identification of metabolic profiles can be decisive for diagnostic and therapeutic management [27,53,54].

For the selection of the number of PCs, in this case, Fig. 5 **(A)** showed that, by using few PCs, high sensitivity is achieved, ensuring the correct classification of most of the target samples (HSIL), but at the expense of specificity, since many CON samples are incorrectly classified, resulte in a higher rate of false negatives. Thus, the adoption of 8 PCs proved ideal because of their ability to balance sensitivity and specificity, which is essential in a clinical context to detect individuals at potential risk of disease progression.

In Fig. 5**(B)**, where HSIL is the target class, it can be observed that most of the calibration (yellow circles) and prediction (yellow diamonds) HSIL samples are correctly positioned within the decision threshold. This indicates that the model efficiently captured the metabolic pattern characteristic of precancerous lesions, considering a value of $\alpha = 0.01$, ensuring that the samples belonging to the HSIL group are accepted within the model. This behavior suggests that the metabolic profile of HSIL is relatively well-defined, allowing statistical modeling based on DD-SIMCA.

Although tests with external validation samples demonstrated the robust specificity of the model, analysis of CON samples revealed considerable dispersion. A few samples were classified within the HSIL group (potential false positives), while others were above the threshold and correctly rejected. The samples classified as false positives do not definitively confirm the presence of lesions in healthy individuals but suggest the possibility of a metabolic overlap between normal profiles and those with precancerous lesions. This overlap may be associated with factors such as individual biological variation, physiological states that share similar metabolic signatures, or the progressive nature of metabolic changes associated with disease progression [55]. Alternatively, these results may reflect the model's possible experimental errors

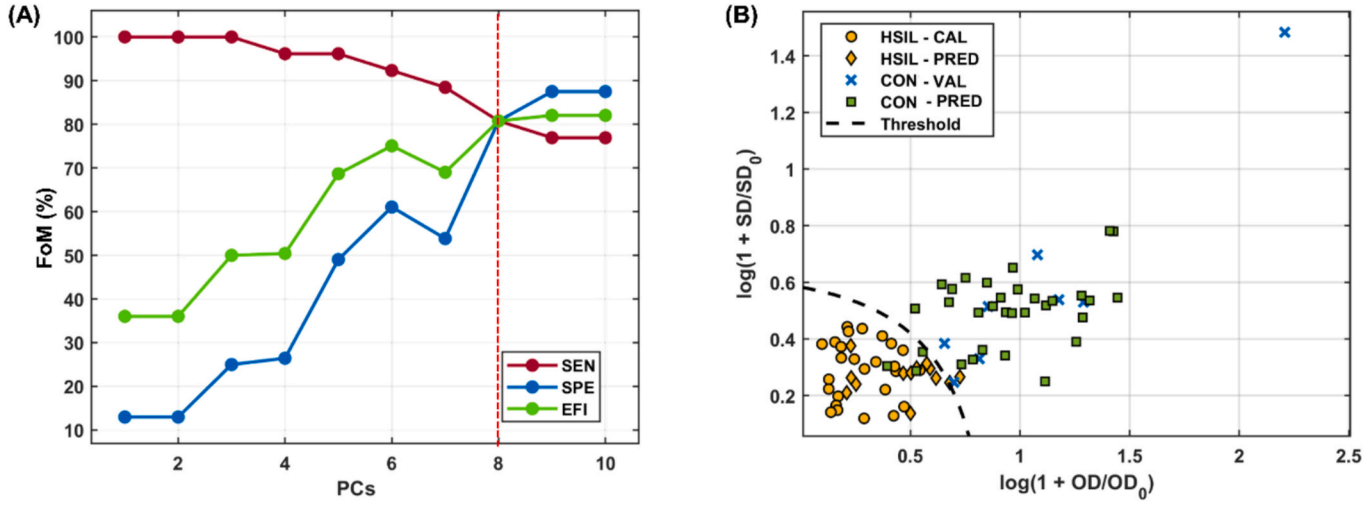


Fig. 5. Results of DD-SIMCA modeling for scenario 2, using the compliant method, in which HSIL was considered as target classes, simulating the clinical application of the method in a screening environment. (A) Evaluation of the model performance when HSIL was defined as the target class, analyzing sensitivity (SEN), specificity (SPE), and overall efficiency (EFI) as a function of the number of principal components (PCs). The model selected to achieve the best balance between sensitivity and specificity used 8 PCs, as indicated by the dashed red line. (B) Samples will be distributed in the DD-SIMCA decision space using the compliant method, considering HSIL as the target class. The x-axis represents the normalized orthogonal distance (OD), and the y-axis represents the normalized score distance (SD). Where: CON − CAL: Samples from the control group used for model calibration; CON − PRED: Samples from the control group used for model prediction; HSIL − VAL: Samples from the HSIL group used in model validation; HSIL − PRED: Samples from the HSIL group used as external samples for prediction testing. The dotted black line represents the statistical decision threshold ($\alpha = 0.01$), above which samples are classified as outside the control model, indicating significant metabolic deviations.

or limitations [28]. In a clinical setting, detecting CON classified as HSIL would serve as an alert for the need for further investigations using additional diagnostic methods to confirm or question the presence of changes suggestive of injury.

The presence CON samples outside the decision threshold reinforce that the model can identify individuals with metabolic profiles significantly different from HSIL. However, the scattered positioning of some CON samples suggests that the model can still be refined by including additional metabolic variables and/or increasing the sample base to better capture the metabolic heterogeneity between groups. Another important aspect is the presence of some HSIL prediction samples close to the decision threshold. This suggests that some individuals within the HSIL group have an intermediate metabolic profile, indicating subgroups within precancerous lesions or variations in the stage of metabolic progression. This observation raises the possibility that different degrees of HSIL may present distinct metabolic signatures. This aspect can be explored in future studies with more robust analytical methods for a more detailed investigation of cervical lesions.

In scenario 3, a smaller number of PCs was needed to obtain more balanced models (Fig. 6 **(A)**). However, similar to what was observed in the previous case, the excessive use of PCs improves specificity but decreases sensitivity, increasing false positives and compromising screening. Thus, 6 PCs were enough to balance sensitivity and specificity. In Fig. 6 **(B)**, where we consider CC as the target class, it can be seen that the calibration (red circles) and prediction (red diamonds) CC samples are well grouped within the target class, demonstrating that the methodology was effective in capturing the characteristic metabolic signature of cervical cancer. The fact that the CC samples were mainly below the statistical threshold indicates that the model could recognize biochemical patterns associated with the tumor stage of the disease. However, the analysis of the CON samples reveals a different behavior from that observed in the previous case (Fig. 5). In this case, more CON samples are positioned outside the model and are correctly rejected. This pattern suggests that the metabolic differences between CON and CC are more pronounced than between CON and HSIL, corroborating the idea that metabolic changes intensify as the disease progresses.

Another relevant point is that some CC samples are located within the area where the CON samples are grouped, which may indicate the existence of common metabolic factors between healthy individuals and patients with cervical cancer. This phenomenon may be partially explained by metabolic interferences associated with lifestyle, individual physiological variations, or transient inflammatory conditions that share similarities with tumor metabolism [55]. Furthermore, the presence of some predictive CC samples positioned close to the threshold raises the hypothesis that, within the cervical cancer group, there may be different metabolic profiles – possibly related to distinct tumor stages, cell subtypes, or variations in the host metabolic response –, suggesting the need for a more refined metabolic stratification in future investigations.

When comparing graphs B in Figs. 5 and 6, it can be seen that the model demonstrated greater efficacy in rejecting CON samples in the CC scenario than in the HSIL scenario, possibly because metabolic alterations in established cancer are more evident than in the precancerous phase. This observation indicates that the metabolic detection of CC may be more accurate. At the same time, the distinction between CON and HSIL presents more significant challenges, reflecting the progressive and gradual nature of metabolic changes in cervical lesions. However, these findings should not be interpreted as a definitive confirmation of the observed phenomena but rather as indications that may reflect biological variability, experimental limitations, or the model itself. Thus, cases in which CC samples overlap with the CON grouping warn of the need for further investigations using additional diagnostic methods to validate and elucidate the real clinical relevance of these metabolic overlaps.

Table 2 presents a quantitative evaluation of the performance of the DD-SIMCA model in classifying the metabolic profiles of cervical lesions, complementing the results previously discussed based on the graphs. The data confirm that the models present high specificity (≥90 %), ensuring that most samples belonging to the external classes are correctly rejected. This result reinforces the ability of the modeling to distinguish individuals with altered metabolic profiles from those considered healthy.

The sensitivity observed in the prediction phase was moderate (55–61.5 %), suggesting the presence of intermediate metabolic profiles, particularly in the HSIL group, which represents a transitional phase before the progression to cervical cancer (CC). The metabolic overlap

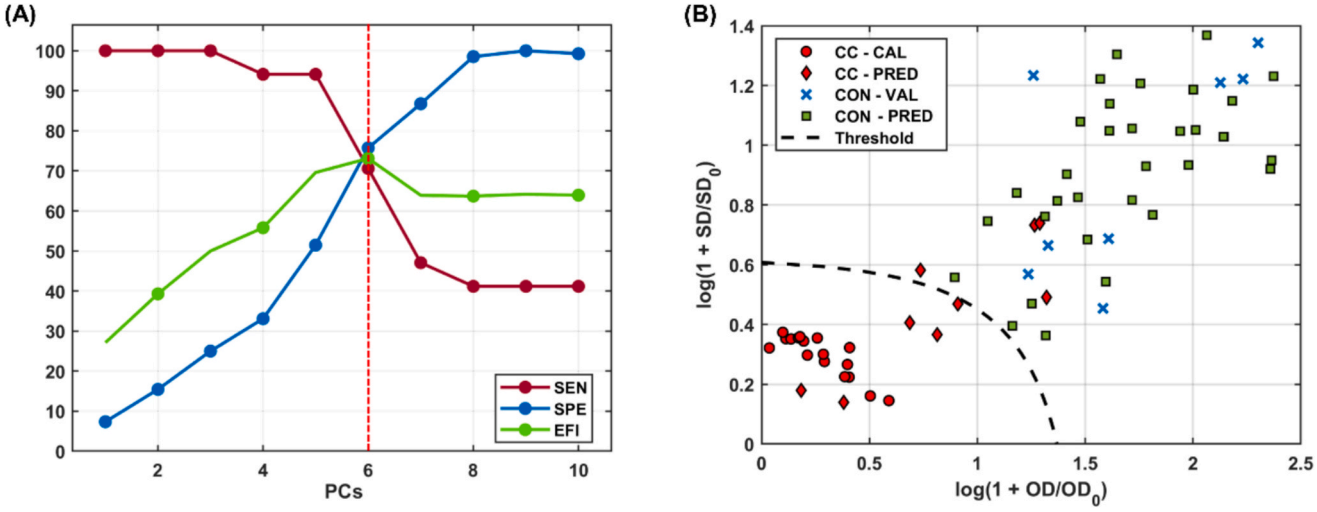


Fig. 6. Results of DD-SIMCA modeling for scenario 3, using the compliant method, in which CC was considered as target classes, simulating the clinical application of the method in a screening environment. (A): Evaluation of model performance as a function of the number of principal components (PCs). The indicators analyzed include sensitivity (SEN), specificity (SPE), and overall efficiency (EFI). The model selected to achieve the best balance between sensitivity and specificity used 6 PCs, as indicated by the dashed red line. (B): Samples will be distributed in the DD-SIMCA decision space, considering CC the target class. The x-axis represents the normalized orthogonal distance (OD), and the y-axis represents the normalized score distance (SD). Where: CON – CAL: Control group samples used for model calibration; CON – VAL: Control group samples used for model validation; CC – VAL: CC group samples used in model validation; CC – PRED: CC group samples used as external samples for prediction testing. The dotted black line represents the statistical decision threshold (α = 0.01), above which samples are classified as outside the CC model, indicating significant metabolic deviations.

Table 2
Performance of the DD-SIMCA model in classifying metabolic profiles of cervical lesions using low-field ^{1}H NMR spectroscopy.

Target Class	External Class	Normalization	N° PCs	α	Data sets	TP	FN	TN	FP	SEN (%)	SPE (%)	ACC (%)	MCC
CON	HSIL and CC	Autoscaling	5	0.01	CAL	27	0	8	0	100	100	100	100
					PRED	8	5	30	4	61.5	90.7	86.3	65.9
HSIL	CON		8		CAL	17	0	8	0	100	100	100	100
					PRED	13	1	29	3	55.0	90.9	91.5	80.2
CC	CON		6		CAL	17	0	8	0	100	100	100	100
					PRED	5	4	32	0	55.5	100	90.2	70.0

between HSIL and CON made it difficult to clearly distinguish between these groups, reflecting the more subtle metabolic changes in early lesions. In contrast, the differentiation between CON and CC was more pronounced, highlighting that the metabolic changes associated with cervical cancer are more evident and easier to identify.

Although the model showed potential for screening, particularly in detecting CC, the moderate sensitivity in the HSIL samples emphasizes the need for improvements in modeling. Such improvements can be achieved through expanding the sample base, incorporating new metabolic variables, and developing more refined modeling strategies. It is worth noting that while sensitivity was moderate in the prediction phase, no external validation sample was incorrectly classified, confirming the model's efficacy in detecting the conditions of interest and its robustness when applied to new datasets. This behavior reinforces confidence in the model's applicability for clinical screening, with a good balance between sensitivity and specificity, especially in detecting cervical cancer.

A relevant parameter in model analysis is the Matthews correlation coefficient (MCC), which provides a balanced metric to assess classification quality. The MCC considers all elements of the confusion matrix (TP, TN, FP, and FN), offering a more robust view of model performance compared to isolated metrics such as accuracy, sensitivity, or specificity. This coefficient ranges from −1 to 1, where values close to 1 indicate strong agreement between model classifications and true sample values, while values close to 0 indicate performance close to chance.

It can be observed that the MCC values were high in all scenarios, ranging from 65.9 % to 80.2 % in the prediction phase (Table 2), which suggests a good correlation between the model predictions and the biological reality of the samples. This index is significant for models such as DD-SIMCA, where the class distribution may be unbalanced, ensuring that the performance is evaluated fairly and without bias towards majority classes. Thus, the MCC reinforces the reliability of the approach, indicating that the model can capture the metabolic differences between the target classes and the controls. However, improvements can be made to reduce the false negative rate.

Overall, the results reinforce that combining low-field ^{1}H NMR spectroscopy with DD-SIMCA is a promising alternative for metabolic screening cervical lesions, enabling early identification of biochemical alterations associated with disease progression. However, methodological advances are still needed to improve the ability to identify individuals in the early stages of the pathological condition, ensuring greater sensitivity without compromising specificity.

The table presents the values of sensitivity (SEN), specificity (SPE), accuracy (ACC), and Matthews correlation coefficient (MCC), in addition to the count of true positives (TP), false negatives (FN), true negatives (TN) and false positives (FP), for the different modeled scenarios. Three target classes were considered: CON (healthy individuals), HSIL (precancerous lesions), and CC (cervical cancer), with the respective external classes used for validation. The model was calibrated and tested separately for each target class, using an optimized number of principal components (PCs) and a significance level of $\alpha = 0.01$. The values obtained in the calibration (CAL) indicate an ideal performance of the model, while the values in the prediction (PRED) reflect the generalization capacity for external samples.

The model's assessment of false positives and negatives is crucial to determine its clinical value. While healthy individuals classified as false negatives are only referred for additional tests, false negatives in HSIL/CC can delay an early diagnosis and compromise therapeutic evolution. To mitigate this risk, a statistical threshold of $\alpha = 0.01$ was adopted, making the classification more rigorous and significantly reducing the probability of missing relevant lesions. Although this strategy slightly increased the number of false negatives in CON, the clinical cost is considerably lower than missing a lesion in a precancerous or cancerous stage.

Furthermore, compared to the Pap smear, which can present false negative rates ranging from 6 % to 60 % [6], the proposed method showed an average accuracy of approximately 91 % in distinguishing between CON and HSIL/CC, suggesting greater consistency and reliability. These results emphasize the potential of the low-field NMR metabolomics strategy as a complementary screening tool since it balances good sensitivity with high specificity and offers less invasiveness compared to traditional methods.

4. Conclusion

The present study evaluated the efficiency of low-field ^{1}H NMR spectroscopy combined with chemometric modeling for detecting cervical lesions, demonstrating great potential in identifying biochemical patterns associated with disease progression. The proposed models achieved high specificity (≥90 %), indicating a robust ability to classify external samples and distinguish healthy individuals (CON) from those with precancerous lesions (HSIL) and cervical cancer (CC).

However, the moderate sensitivity (55–61.5 %) in the prediction phase indicates that some HSIL and CC samples were classified as CON, possibly due to intermediate metabolic profiles in the HSIL group, reflecting its transitional stage before evolving into CC. This phenomenon was corroborated by the PCA, where PC1 discriminated CC from the other groups, while HSIL and CON showed overlap, suggesting shared metabolic signatures. The more subtle variations, evidenced by PC2 and PC3, reflect the importance of spectral regions indicating the presence of metabolites such as glycine ($\delta = 3.61$ and 3.56 ppm), choline ($\delta = 3.19$ and 3.14 ppm), N-acetylglycoproteins (NAG, $\delta = 2.00$ ppm), lactate ($\delta = 1.26$ ppm), and lipids ($\delta = 0.96$ ppm) in the differentiation of metabolic profiles.

Although the technique has limitations in the precise identification of metabolites due to the spectral resolution of low-field NMR, the analysis of the loading values allowed us to suggest the most likely compounds, based on the initial peak assignments. This provided a solid foundation for interpreting the variations in the metabolic profiles and better understanding the potential metabolic markers associated with the observed behaviors. The analysis of the Matthews correlation coefficient (MCC) reinforced the reliability of the models by evaluating their performance in a balanced manner. However, the presence of CON samples within the HSIL and CC model spaces suggests that individual metabolic variations or localized inflammations may influence the results, which could be further investigated in future studies. These studies, using more advanced techniques, could validate the observations made and investigate the action of the identified compounds with greater precision

Thus, the results reinforce the potential of the new proposed

approach as a robust tool for metabolic screening, allowing the early identification of individuals at increased risk of cervical lesions. In clinical terms, this approach shows promise as a complementary screening tool, as it offers quantitative assessment, high specificity, and less invasiveness compared to the standard method, such as the Pap smear, and can be applied on a large scale with greater comfort to the patient, consolidating the use of low-field MRI as a viable and minimally invasive alternative in clinical practice.

Data availability statement

No data was used for the research described in the article.

CRediT authorship contribution statement

Luana Quadros de Souza Leão: Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jelmir Craveiro de Andrade:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Giovanna Melo Marques:** Methodology, Data curation, Investigation, Formal analysis. **Cristiane Cunha Guimarães:** Methodology. **Rosyana de Fátima Vieira de Albuquerque:** Writing – review & editing, Methodology, Data curation. **Alessandra Silva e Silva:** Methodology, Investigation, Data curation. **Kamila Pereira de Araujo:** Methodology, Data curation, Conceptualization. **Mikele Praia de Oliveira:** Visualization. **Anderson Ferreira Gonçalves:** Visualization. **Higino Felipe Figueiredo:** Visualization. **Daniel Lourenço Lira:** Visualization. **Marina Amaral Alves:** Methodology, Investigation, Formal analysis, Conceptualization. **Carlos Adam Conte-Junior:** Writing – original draft, Supervision, Project administration, Data curation, Conceptualization. **Priscila Ferreira de Aquino:** Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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